

Chris Eugene

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PROFESSIONAL SUMMARY

Electrical and Computer Engineering graduate (M.S., Saint Louis University, May 2026, GPA: 3.90/4.0) with a versatile technical background spanning machine learning, computer vision, semiconductor fabrication, power systems design, embedded systems, optical communication, and renewable energy engineering. Proficient in Python and TypeScript; experienced in end-to-end project delivery, ISO 9001 compliance, data analysis, and cross-functional collaboration. Proven ability to design, build, and communicate complex engineering systems across hardware, software, and research domains. Strong analytical problem-solver with rigorous documentation discipline and a fast-learning mindset. Open to engineering, research, data, and technology roles across multiple industries.

TECHNICAL SKILLS

Programming & ML: Python (proficient — NumPy, Pandas, Scikit-learn, TensorFlow, OpenCV, data pipelines, automation), TypeScript (proficient), C# (familiar), C++ (familiar), SQL (basic), Git

AI & GenAI: Deep learning, computer vision, LLMs (GPT, BERT — conceptual), prompt engineering, embeddings, generative models (GANs, VAEs, Diffusion), LangChain (familiar), Hugging Face (familiar), OpenAI API (familiar), RAG pipelines (familiar), Azure AI / AWS (familiar)

Semiconductor & Materials: CVD thin film growth (MoS_2 /graphene on SiO_2/Si), AFM direct-write patterning, thermal evaporation, magnetron sputtering, PMMA-assisted wet transfer, photolithographic contact patterning, Raman spectroscopy, SEM, AFM, UV-Vis, photoluminescence spectroscopy

Electrical & Power Systems: AC/DC power systems, single & three-phase power, transformer design (CT/VT, step-up/down), protection relay schemes, AVR, SCADA, embedded systems, relay logic, sequential control, circuit analysis, power factor & reactive loading

Design & Documentation: Electrical one-line diagrams, control schematics, BOM generation, process flow documentation, SOP development, ISO 9001 / ISO 17100 compliance, CAPA, root cause analysis, audit-ready record keeping, AutoCAD (familiar), MATLAB (familiar)

Optical & Networking: Laser-based Li-Fi simulation, OOK modulation, geometric beam propagation, TCP/IP, LAN/WAN, EtherNet/IP, PROFIBUS, PROFINET (academic)

Tools & Practices: Microsoft Excel (advanced), Tableau (familiar), Power BI (familiar), Git, Agile/Scrum (familiar), cross-functional team collaboration, technical presentations, global team alignment, training & knowledge development

ACADEMIC PROJECTS

MoS_2 /Graphene Heterostructures for Scalable Solar Energy (Team KFC) 2026

Saint Louis University — St. Louis, MO — ECE5930: Materials, Characterizations & Device Applications

- Co-fabricated high-crystalline MoS_2 /graphene heterostructures on SiO_2/Si substrates using mask-free AFM direct-write precursor patterning followed by CVD sulfurization (600–900°C, S/Ar ambient).
- Confirmed few-layer MoS_2 formation via Raman spectroscopy ($E^*_g \approx 385 \text{ cm}^{-1}$, $A^*_g \approx 405 \text{ cm}^{-1}$, $\Delta\omega \approx 20 \text{ cm}^{-1}$) and characterized triangular domain morphology and grain boundary behavior via SEM.
- Optimized CVD growth parameters iteratively to achieve high-crystallinity domains for optoelectronic device integration; documented process flows and characterization data in structured technical reports.

Plasmon-Enhanced Graphene/ MoS_2 /Ag Heterostructure Photodetector 2026

Saint Louis University — St. Louis, MO — Nanomaterial Research

- Designed a vertical photodetector stack leveraging LSPR of nanostructured Ag to enhance electromagnetic field confinement and optical absorption in MoS_2 .
- Developed full fabrication process flow: Ag nanostructure formation (thermal evaporation/magnetron sputtering) → MoS_2 CVD growth → PMMA-assisted graphene wet transfer → photolithographic metal contact patterning.
- Specified device performance targets (responsivity $>0.5 \text{ A/W}$, detectivity $>10^{12} \text{ Jones}$, response time $<10 \text{ ms}$) and planned full characterization suite: UV-Vis, Raman, AFM, SEM, and PL spectroscopy.

Laser-Based Li-Fi Communication with Automatic Beam Steering

2025

Saint Louis University — St. Louis, MO — CSCI 4/5550 Computer Networks

- Designed and built a Python/Pygame simulation of a laser-based Li-Fi system featuring a fixed transmitter, movable receiver, and intelligent beam splitter with automatic steering using geometric beam propagation and line-rectangle collision detection.
- Implemented OOK binary data transmission — 8-bit input displayed at receiver only when optical link established via direct or reflected beam path; documented system architecture, protocol design, and performance trade-offs.

Object Localization using Deep Learning Models

2025

Saint Louis University — St. Louis, MO

- Built an end-to-end computer vision pipeline in Python using TensorFlow and OpenCV — data preprocessing, feature engineering, model architecture design, training, hyperparameter tuning, validation, performance benchmarking, and deployment.
- Wrote and debugged complex detection and localization algorithms; evaluated model outputs for accuracy and localization quality; produced structured analysis reports and documentation throughout development.

Micro Hydro-Electric Power Generation — Blue Nile River, Ethiopia

2026

Saint Louis University — St. Louis, MO — Advanced Energy Systems

- Designed a complete 100 kW run-of-river micro-hydroelectric system across civil, mechanical, electrical, and commissioning disciplines on a 24-month phased schedule with \$455.5K CAPEX and full NPV analysis.
- Specified electrical distribution: 100 kVA 3-phase synchronous generator, 400V→11kV step-up transformer, AVR, protection relays, SCADA panel, 2km MV distribution line, and 477-household metering; generated full project BOM and commissioning documentation.

Automated Energy-Saving System with Sequential Response Control

2022–2023

St. Joseph's College of Engineering — Chennai, India

- Designed and prototyped a hardware-based automated energy management system using sequential switching control logic, relay feedback, and real-time sensor monitoring.
- Executed functional testing, iterative system optimization, and documented architecture, logic diagrams, and performance results in a structured technical report.

PROFESSIONAL EXPERIENCE

Teaching Assistant — Energy Conversions

Jan 2026 – May 2026

Saint Louis University — St. Louis, MO

- Designed and delivered lab instruction on AC/DC power systems, electrical-mechanical energy conversion, and transformer operation for 20+ undergraduate students.
- Tracked individual progress across 20+ learners, identified performance gaps, implemented targeted corrective interventions, and maintained 100% on-time grading throughout the semester.
- Developed standardized assessment rubrics, instructional materials, and knowledge-sharing resources; collaborated with the course instructor on curriculum adjustments and student performance reporting.

Teaching Assistant — College Physics

Aug 2025 – May 2026

Saint Louis University — St. Louis, MO

- Led structured lab sessions in mechanics, electricity, and optics for 25+ undergraduate students; provided individualized tutoring on electrical fundamentals and quantitative problem-solving.
- Diagnosed and resolved lab equipment failures in real time using root cause analysis; maintained 100% on-time grading across two full semesters.
- Delivered instruction on wave phenomena, electromagnetic fundamentals, and optical physics underpinning semiconductor device behavior.

Quality Assurance Specialist

Oct 2023 – Jul 2024

TransPerfect — Chennai, India

- Enforced ISO 9001 and ISO 17100 compliance across global operations; developed and maintained SOPs for 10+ concurrent workflows.

- Conducted root cause analysis on operational failures and designed CAPA plans, reducing repeat incidents and improving process reliability across global teams.
- Managed weekly cost tracking, budget reporting, and compliance documentation across multiple concurrent accounts; met all reporting deadlines on schedule.
- Collaborated with worldwide teams to benchmark documentation standards and drive process alignment; onboarded and trained 10+ new team members on complex technical processes.

Networking & IT Support Engineering Intern

Aug 2023 – Oct 2023

Mercedes-Benz Titanium Motors — Chennai, India

- Diagnosed and resolved network and system faults across dealership operations using TCP/IP analysis and systematic troubleshooting; maintained organized incident logs and configuration documentation.
- Configured and maintained LAN infrastructure and device connectivity; communicated technical findings clearly to non-technical staff and management.

EDUCATION

Master of Science — Electrical and Computer Engineering May 2026

Saint Louis University — St. Louis, MO

GPA: 3.90 / 4.0 | Relevant Coursework: Deep Learning & Computer Vision, Materials & Device Characterization (ECE5930), Energy Conversions, Power Systems, Embedded Systems, Computer Networks, Algorithms

Bachelor of Science — Electrical and Electronics Engineering May 2023

St. Joseph's College of Engineering — Chennai, India

GPA: 3.67 / 4.0 | Relevant Coursework: Semiconductor Devices, Power Systems, Electrical Machines, Control Systems, Electronic Circuits, Microprocessors, Digital Electronics, Engineering Materials